

RESTAURANT POS SYSTEM

Fullstack Project Directory Blueprint & Architecture Reference Guide

Project Name:	Restaurant Point of Sale System	Generated Date:	August 11, 2026
Root Directory:	c:\Restaurant Project	Tech Stack:	Java 17 Spring Boot + React 18 SPA
Total Directories:	135 Sub-directories	Total Source Files:	295 Source & Asset Files

1. System Architecture Overview

The project is structured into four primary modules: the Spring Boot Java Backend API, the React Single Page Web Application (website), UI Wireframes, and root API & database documentation.

Component Module	Technology Stack	Functional Responsibility & Domain Scope
backend/	Java 17, Spring Boot 3, JPA, Hibernate, Security JWT	Core REST microservices: Billing, Inventory, Staff, Order Processing, Kitchen Display System (KDS), Delivery Partners, and Analytics.
website/	React 18, Vite, Vanilla CSS, Axios, WebSockets	Front-of-house & Back-of-house Web POS interfaces with role-based navigation (Admin, Captain, Cashier, Kitchen, Manager).
UI Design/	Visual JPEG Wireframes & Interface Mockups	Screen designs categorized by operational role (Admin, Cashier, Delivery, Kitchen, Manager, Waiter).
Root Specifications	PDF & Text Documentation	API Endpoint Testing Guides and Relational Database Design Schema Reports.

2. Project Root Directory View

Top-level directory layout showing main sub-projects and documentation files.

```
Restaurant Project/
|-- UI Design/
| |-- Admin/
| |-- Cashier/
| |-- Delivery/
| |-- Kitchen Staff/
| |-- Manager/
| \-- Waiter/
|-- backend/
| |-- src/
| |-- HELP.md
| |-- mvnw
| |-- mvnw.cmd
| \-- pom.xml
|-- website/
| |-- public/
```

```
| |-- src/
| |-- README.md
| |-- index.html
| |-- package-lock.json
| |-- package.json
| \-- vite.config.js
|-- API_ENDPOINTS_TESTING_GUIDE.txt
|-- Restaurant_POS_API_Endpoints_Guide.pdf
|-- Restaurant_POS_Database_Design_Report.pdf
\-- Restaurant_POS_Folder_Structure.pdf
```

3. Backend Service Architecture (`backend/`)

Spring Boot 3 application directory tree: Domain entities, Controllers, DTOs, JPA Repositories, Security, and Integration Test suites.

```
backend/
|-- src/
| |-- main/
| | |-- java/
| | | |-- com/
| | | | |-- example/
| | | | |-- backend/
| | | | |-- admin/
| | | | |-- controller/
| | | | | |-- AdminController.java
| | | | |-- dto/
| | | | | |-- AdminRequest.java
| | | | | |-- AdminResponse.java
| | | | |-- entity/
| | | | | |-- Admin.java
| | | | |-- repository/
| | | | | |-- AdminRepository.java
| | | | |-- service/
| | | | | |-- AdminService.java
| | | |-- auth/
| | | | |-- controller/
| | | | | |-- AuthController.java
| | | | |-- dto/
| | | | | |-- LoginRequest.java
| | | | | |-- LoginResponse.java
| | | | |-- service/
| | | | | |-- AuthService.java
| | | |-- bill/
| | | | |-- controller/
| | | | | |-- BillController.java
| | | | |-- dto/
| | | | | |-- BillPaymentSplitRequest.java
| | | | | |-- CashDrawerReconciliation.java
| | | | | |-- SettleBillRequest.java
| | | | | |-- SplitBillByItemsRequest.java
| | | | |-- entity/
| | | | | |-- Bill.java
| | | | | |-- BillPayment.java
| | | | |-- repository/
| | | | | |-- BillPaymentRepository.java
```

```
| | | | \-- BillRepository.java
| | | | \-- service/
| | | | \-- BillService.java
| | | | -- chain/
| | | | -- controller/
| | | | \-- ChainController.java
| | | | -- dto/
| | | | |-- ConsolidatedChainReportResponse.java
| | | | |-- OutletComparisonResponse.java
| | | | |-- OverrideMenuRequest.java
| | | | \-- PushMenuRequest.java
| | | | -- entity/
| | | | |-- Brand.java
| | | | |-- BrandPromotion.java
| | | | |-- CorporateNotification.java
| | | | |-- Outlet.java
| | | | \-- OutletMenuOverride.java
| | | | -- repository/
| | | | |-- BrandPromotionRepository.java
| | | | |-- BrandRepository.java
| | | | |-- CorporateNotificationRepository.java
| | | | |-- OutletMenuOverrideRepository.java
| | | | \-- OutletRepository.java
| | | | \-- service/
| | | | \-- ChainService.java
| | | | -- common/
| | | | \-- service/
| | | | \-- EmailService.java
| | | | -- config/
| | | | |-- DataSeeder.java
| | | | \-- WebSocketConfig.java
| | | | -- customer/
| | | | |-- controller/
| | | | |-- CustomerAppController.java
| | | | |-- CustomerController.java
| | | | |-- dto/
| | | | |-- CustomerAnalyticsResponse.java
| | | | |-- CustomerFeedbackRequest.java
| | | | |-- CustomerOrderTrackingResponse.java
| | | | |-- CustomerRequest.java
| | | | |-- CustomerResponse.java
| | | | |-- DirectOrderRequest.java
| | | | \-- OtpVerifyRequest.java
| | | | -- entity/
| | | | |-- Customer.java
| | | | |-- CustomerVisit.java
| | | | -- repository/
| | | | |-- CustomerRepository.java
| | | | |-- CustomerVisitRepository.java
| | | | \-- service/
| | | | \-- CustomerService.java
| | | | -- delivery/
| | | | |-- controller/
| | | | |-- DeliveryManagementController.java
| | | | |-- DeliveryPartnerController.java
| | | | |-- dto/
```

```
| | | | |-- CompleteDeliveryRequest.java
| | | | |-- DeliveryAssignmentResponse.java
| | | | |-- DeliveryPartnerRequest.java
| | | | |-- DeliveryPartnerResponse.java
| | | | |-- DeliveryPinLoginRequest.java
| | | | |-- DeliveryPinLoginResponse.java
| | | | \-- LocationUpdateRequest.java
| | | | |-- entity/
| | | | | |-- AssignmentStatus.java
| | | | | |-- DeliveryAssignment.java
| | | | | |-- DeliveryPartner.java
| | | | | \-- DeliveryPartnerStatus.java
| | | | |-- repository/
| | | | | |-- DeliveryAssignmentRepository.java
| | | | | \-- DeliveryPartnerRepository.java
| | | | \-- service/
| | | | | \-- DeliveryService.java
| | | | |-- exception/
| | | | | |-- BillAlreadySettledException.java
| | | | | |-- GlobalExceptionHandler.java
| | | | | |-- IngredientStockException.java
| | | | | |-- KOTAlreadyFiredException.java
| | | | | |-- ManagerPINRequiredException.java
| | | | | \-- TableOccupiedException.java
| | | | |-- inventory/
| | | | | |-- controller/
| | | | | | \-- InventoryController.java
| | | | | |-- dto/
| | | | | | |-- CogsReportResponse.java
| | | | | | |-- DailyStockCloseRequest.java
| | | | | | |-- DailyStockSetupRequest.java
| | | | | | |-- PurchaseOrderItemRequest.java
| | | | | | |-- PurchaseOrderRequest.java
| | | | | | \-- SupplierRequest.java
| | | | | |-- entity/
| | | | | | |-- DailyStockSheet.java
| | | | | | |-- InventoryItem.java
| | | | | | |-- PerishableBatch.java
| | | | | | |-- PurchaseOrder.java
| | | | | | |-- PurchaseOrderItem.java
| | | | | | |-- PurchaseOrderStatus.java
| | | | | | |-- Recipe.java
| | | | | | \-- Supplier.java
| | | | | |-- repository/
| | | | | | |-- DailyStockSheetRepository.java
| | | | | | |-- InventoryItemRepository.java
| | | | | | |-- PerishableBatchRepository.java
| | | | | | |-- PurchaseOrderItemRepository.java
| | | | | | |-- PurchaseOrderRepository.java
| | | | | | |-- RecipeRepository.java
| | | | | | \-- SupplierRepository.java
| | | | | \-- service/
| | | | | | \-- InventoryService.java
| | | | |-- menu/
| | | | | |-- controller/
| | | | | | \-- KitchenStationController.java
```

```
| | | | \-- MenuController.java
| | | | |-- entity/
| | | | |-- Category.java
| | | | |-- ComboComponent.java
| | | | |-- KitchenStation.java
| | | | |-- MenuItem.java
| | | | |-- ModifierGroup.java
| | | | \-- ModifierOption.java
| | | | |-- repository/
| | | | |-- CategoryRepository.java
| | | | |-- ComboComponentRepository.java
| | | | |-- KitchenStationRepository.java
| | | | |-- MenuItemRepository.java
| | | | |-- ModifierGroupRepository.java
| | | | \-- ModifierOptionRepository.java
| | | | \-- service/
| | | | |-- MenuService.java
| | | | \-- QrCodeService.java
| | | |-- order/
| | | | |-- controller/
| | | | |-- KdsController.java
| | | | \-- OrderController.java
| | | | |-- dto/
| | | | |-- KdsItemResponse.java
| | | | |-- KdsPerformanceResponse.java
| | | | |-- KdsSyncRequest.java
| | | | |-- OrderItemSplitRequest.java
| | | | \-- OrderRequest.java
| | | | |-- entity/
| | | | |-- Order.java
| | | | |-- OrderItem.java
| | | | |-- OrderItemStatus.java
| | | | |-- OrderStatus.java
| | | | |-- OrderTimelineEvent.java
| | | | \-- OrderType.java
| | | | |-- repository/
| | | | |-- OrderItemRepository.java
| | | | |-- OrderRepository.java
| | | | \-- OrderTimelineRepository.java
| | | | \-- service/
| | | | |-- KdsService.java
| | | | |-- OrderEscalationScheduler.java
| | | | \-- OrderService.java
| | | |-- report/
| | | | |-- controller/
| | | | |-- AnalyticsController.java
| | | | |-- dto/
| | | | |-- AggregatorAnalyticsResponse.java
| | | | |-- CustomerAnalyticsReportResponse.java
| | | | |-- DailySalesSummaryResponse.java
| | | | |-- DeliveryAnalyticsResponse.java
| | | | |-- ItemPerformanceResponse.java
| | | | |-- ItemSalesDto.java
| | | | |-- TableAnalyticsResponse.java
| | | | \-- XReportResponse.java
| | | | |-- entity/
```

```
| | | | \-- DayEndReport.java
| | | | |-- repository/
| | | | \-- DayEndReportRepository.java
| | | | \-- service/
| | | | \-- AnalyticsService.java
| | | | |-- reservation/
| | | | |-- controller/
| | | | \-- ReservationController.java
| | | | |-- dto/
| | | | \-- ReservationAnalyticsResponse.java
| | | | |-- entity/
| | | | |-- Reservation.java
| | | | |-- ReservationStatus.java
| | | | |-- repository/
| | | | \-- ReservationRepository.java
| | | | \-- service/
| | | | |-- ReservationScheduler.java
| | | | |-- ReservationService.java
| | | | |-- security/
| | | | |-- CustomUserDetailService.java
| | | | |-- JwtAuthenticationFilter.java
| | | | |-- JwtUtil.java
| | | | \-- SecurityConfig.java
| | | | |-- staff/
| | | | |-- controller/
| | | | |-- StaffController.java
| | | | |-- dto/
| | | | |-- PinLoginRequest.java
| | | | |-- PinLoginResponse.java
| | | | |-- ShiftAssignmentRequest.java
| | | | |-- StaffPerformanceResponse.java
| | | | |-- StaffRequest.java
| | | | |-- StaffResponse.java
| | | | |-- entity/
| | | | |-- AttendanceLog.java
| | | | |-- AuditLog.java
| | | | |-- Shift.java
| | | | |-- ShiftAssignment.java
| | | | |-- Staff.java
| | | | |-- StaffRole.java
| | | | |-- repository/
| | | | |-- AttendanceLogRepository.java
| | | | |-- AuditLogRepository.java
| | | | |-- ShiftAssignmentRepository.java
| | | | |-- ShiftRepository.java
| | | | |-- StaffRepository.java
| | | | \-- service/
| | | | |-- StaffService.java
| | | | |-- table/
| | | | |-- controller/
| | | | |-- TableController.java
| | | | |-- WaitlistController.java
| | | | |-- entity/
| | | | |-- RestaurantTable.java
| | | | |-- TableStatus.java
| | | | |-- WaitlistEntry.java
```

```
| | | | \-- WaitlistStatus.java
| | | | |-- repository/
| | | | |-- RestaurantTableRepository.java
| | | | |-- WaitlistRepository.java
| | | | |-- service/
| | | | |-- TableService.java
| | | | |-- WaitlistService.java
| | | \-- BackendApplication.java
| | \-- resources/
| | |-- static/
| | |-- templates/
| | \-- application.properties
| \-- test/
| \-- java/
| \-- com/
| \-- example/
| \-- backend/
| |-- bill/
| | \-- PaymentProcessingIntegrationTests.java
| |-- chain/
| | \-- MultiOutletChainIntegrationTests.java
| |-- customer/
| | \-- CustomerCrmIntegrationTests.java
| |-- delivery/
| | \-- DirectCustomerAndDeliveryPartnerIntegrationTests.java
| |-- inventory/
| | \-- InventoryManagementIntegrationTests.java
| |-- order/
| | |-- KitchenDisplaySystemIntegrationTests.java
| | \-- OrderManagementIntegrationTests.java
| |-- report/
| | \-- AnalyticsReportingIntegrationTests.java
| |-- reservation/
| | \-- ReservationManagementIntegrationTests.java
| |-- staff/
| | \-- StaffManagementIntegrationTests.java
| |-- table/
| | \-- TableManagementIntegrationTests.java
| \-- BackendApplicationTests.java
|-- HELP.md
|-- mvnw
|-- mvnw.cmd
\-- pom.xml
```

4. Frontend Web Application (`website/`)

React 18 + Vite SPA directory layout: Components, Contexts, Role Dashboards, Services, Assets, and Configuration files.

```
website/
|-- public/
| |-- favicon.svg
| |-- icons.svg
|-- src/
| |-- assets/
| | |-- hero.png
```

```
| | |-- react.svg
| | |-- vite.svg
| |-- components/
| | |-- chain/
| | | |-- ChainOutletManager.jsx
| | |-- common/
| | | |-- HotkeyHandler.jsx
| | |-- discounts/
| | | |-- DiscountPromotionsManager.jsx
| | |-- floor/
| | | |-- FloorPlanManager.jsx
| | |-- inventory/
| | | |-- InventoryStockManager.jsx
| | |-- kitchen/
| | | |-- KitchenStationManager.jsx
| | |-- layout/
| | | |-- Header.jsx
| | | |-- Navbar.jsx
| | | |-- RoleGuard.jsx
| | | |-- Sidebar.jsx
| | |-- menu/
| | | |-- MenuCatalogManager.jsx
| | |-- reports/
| | | |-- ReportsAnalyticsManager.jsx
| | |-- reservation/
| | | |-- ReservationWaitlistManager.jsx
| |-- staff/
| | |-- StaffShiftManager.jsx
| |-- context/
| | |-- AuthContext.jsx
| |-- pages/
| | |-- admin/
| | | |-- AdminDashboard.jsx
| | |-- captain/
| | | |-- CaptainDashboard.jsx
| | |-- cashier/
| | | |-- CashierDashboard.jsx
| | |-- kitchen/
| | | |-- KitchenDashboard.jsx
| | |-- manager/
| | | |-- ManagerDashboard.jsx
| | |-- Login.jsx
| |-- services/
| | |-- apiClient.js
| | |-- audioService.js
| | |-- authService.js
| | |-- websocketService.js
| |-- App.css
| |-- App.jsx
| |-- index.css
| |-- main.jsx
|-- README.md
|-- index.html
|-- package-lock.json
|-- package.json
-- vite.config.js
```

5. UI Design Wireframes (`UI Design/`)

Categorized visual design wireframe mockups for each operational role.

```
UI Design/  
|-- Admin/  
| |-- visily-dashboard.jpg  
| |-- visily-floor-plan.jpg  
| |-- visily-menu.jpg  
| |-- visily-mobile-dashboard.jpg  
| |-- visily-reports.jpg  
| |-- visily-settings.jpg  
| \-- visily-users.jpg  
|-- Cashier/  
| |-- visily-billing-management.jpg  
| |-- visily-cashier-dashboard.jpg  
| |-- visily-day-closing.jpg  
| |-- visily-invoice-generation.jpg  
| |-- visily-navigation-&-profile.jpg  
| |-- visily-payment-processing.jpg  
| |-- visily-receipt-management.jpg  
| |-- visily-split-bill.jpg  
| \-- visily-transaction-history.jpg  
|-- Delivery/  
| |-- visily-assigned-orders.jpg  
| |-- visily-components-&-structure.jpg  
| |-- visily-customer-info.jpg  
| |-- visily-delivery-confirmation.jpg  
| |-- visily-delivery-dashboard.jpg  
| |-- visily-delivery-history.jpg  
| |-- visily-live-navigation.jpg  
| |-- visily-order-details-&-pickup.jpg  
| |-- visily-order-status-workflow.jpg  
| |-- visily-performance-&-analytics.jpg  
| \-- visily-profile-&-support.jpg  
|-- Kitchen Staff/  
| |-- visily-completed-orders.jpg  
| |-- visily-kitchen-dashboard.jpg  
| |-- visily-kitchen-queue.jpg  
| |-- visily-pending-orders.jpg  
| |-- visily-preparing-orders.jpg  
| |-- visily-ready-orders.jpg  
| \-- visily-staff-profile.jpg  
|-- Manager/  
| |-- visily-dashboard (1).jpg  
| |-- visily-discount-approvals.jpg  
| |-- visily-edit-menu-item.jpg  
| |-- visily-inventory-management.jpg  
| |-- visily-menu-management.jpg  
| |-- visily-notifications-center.jpg  
| |-- visily-sales-reports.jpg  
| \-- visily-staff-management.jpg  
\-- Waiter/  
|-- visily-customer-profile.jpg
```

```
|-- visily-floor-dashboard.jpg  
|-- visily-kitchen-tickets.jpg  
|-- visily-modify-order.jpg  
|-- visily-order-cart.jpg  
|-- visily-order-taking.jpg  
|-- visily-reservations.jpg  
\-- visily-table-transfer.jpg
```